

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)*

Assessment Tool Weightage:20%

Analyze the given problem by comparing available techniques against the specified constraints and requirements. Evaluate and justify your decision using logical reasoning, clearly indicating how the chosen approach satisfies the conditions.

Given Constraints

- Latency must be less than or equal to 100 ms.
- Accuracy must be greater than or equal to 85%.
- Hardware supports only moderate computation.

Model	Accuracy	Computation/Latency	Evaluation
Viola-Jones	78%	Low latency	Does not satisfy accuracy
YOLO	92%	Moderate	Satisfies all constraints
Deep Learning	95%	High	Does not satisfy hardware/latency constraint

Comparison: Viola-Jones is very fast but its accuracy is only 78%, which is below the required 85%. Deep Learning provides the highest accuracy but requires high

computation and has high latency. YOLO provides 92% accuracy with moderate computation.

Conclusion: YOLO should be selected because it satisfies all the given constraints and provides a good balance between accuracy, latency, and computational requirements.

2. Autonomous Vehicle Detection Logic

Given Constraints

- Accuracy must be greater than 90%.
- Response time must be less than 50 ms.

Method	Accuracy	Response Time	Evaluation
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A	88%	30 ms	Accuracy is insufficient
B	93%	70 ms	Response time is too high
C	91%	45 ms	Satisfies both constraints

Comparison: Method A is fast but does not provide sufficient accuracy. Method B provides high accuracy but takes 70 ms, which exceeds the 50 ms limit. Method C provides 91% accuracy and responds within 45 ms.

Conclusion: Method C should be selected because it satisfies both the accuracy and response-time requirements.

3. Motion Analysis Decision Problem

Approach	Advantage	Limitation
Optical Flow	Accurate and detailed	Noisy in dense traffic
Motion Estimation	Stable	Less detailed

Comparison: Optical Flow provides detailed information about motion direction, but in dense traffic, multiple vehicles and overlapping movements can produce noisy vectors. Motion Estimation provides more stable results but may not capture fine motion details.

Evaluation: In high-density traffic, stability is more important because noisy motion information can lead to incorrect decisions.

Conclusion: Motion Estimation is more suitable for dense traffic because it provides stable motion tracking. A hybrid system can use Optical Flow in low-density traffic and Motion Estimation in high-density traffic.

4. Background Modeling Evaluation

Scenario	GMM Accuracy	Required Accuracy
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Static	90%	80%
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Dynamic	65%	80%
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Comparison: GMM performs well in static scenes with 90% accuracy. However, its accuracy falls to 65% in dynamic scenes.

Evaluation: The system requires at least 80% accuracy in every scenario. Since GMM achieves only 65% in dynamic scenes, it does not satisfy the overall requirement.

Conclusion: GMM alone should not be selected. It should be combined with techniques such as Optical Flow, shadow removal, adaptive background updating, or deep-learning object detection.

5. Depth Estimation Strategy

Method	Accuracy	Camera Requirement	Evaluation
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Stereo Vision	High	Two cameras	Not feasible
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Structure from Motion	Moderate	One camera	Feasible
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Comparison: Stereo Vision provides accurate depth but requires two cameras. Structure from Motion provides moderate depth accuracy using only one camera.

Evaluation: The system has only one camera, and moderate accuracy is acceptable. Therefore, Stereo Vision cannot satisfy the hardware constraint, while Structure from Motion can.

Conclusion: **Structure from Motion should be selected** because it works with a single camera and provides acceptable depth accuracy.

6. Retail Motion Tracking Logic

Method	Low Crowd Density	High Crowd Density
Optical Flow	Accurate	Noisy
Motion Models	Less detailed	Stable

Comparison: Optical Flow is accurate when customers are separated and individual movements can be identified. In crowded environments, overlapping movements cause noise. Motion Models are more stable in dense crowds but provide less detailed motion information.

Evaluation: Since the system must work effectively in both low- and high-density situations, using only one method is not ideal.

Conclusion: **A hybrid approach is the best solution.** Optical Flow should be used in low-density areas, while Motion Models should be used in high-density areas.

7. Defect Detection System Decision

Requirements

- Detect subtle defects with more than 95% accuracy.
- Adapt to product variations.

Approach	Speed	Accuracy/Adaptability
Traditional Methods	Fast	Less accurate and less adaptive
Deep Learning	Slower	Highly accurate and adaptive

Comparison: Traditional computer vision methods are fast but may fail when product appearance changes. Deep Learning can learn complex defect patterns and handle variations better when trained with a diverse dataset.

Evaluation: The requirement demands very high accuracy and adaptability. Therefore, traditional methods are not sufficient.

Conclusion: Deep Learning should be selected because it is better suited for detecting subtle defects and adapting to product variations.

8. AR Motion Tracking Constraint Problem

Requirements

- Tracking drift must be less than 5%.
- System must run on mobile hardware.

Method	Tracking Detail	Computation
Optical Flow	High	High
Motion Estimation	Moderate	Low

Comparison: Optical Flow provides detailed tracking but requires significant computation. This can be difficult for mobile hardware. Motion Estimation is less detailed but computationally efficient.

Evaluation: Since the system must run on mobile hardware, computational efficiency is important. Motion Estimation is therefore more suitable, provided its drift can be controlled below 5%.

Conclusion: Motion Estimation should be selected for mobile AR. Sensor fusion and periodic correction can be used to reduce drift.

9. Mobile Face Detection Trade-off

Requirements

- Real-time operation.
- Accuracy must be at least 85%.

Method	Accuracy	Speed	Evaluation
Viola-Jones	80%	Fast	Accuracy is insufficient
Deep Learning	95%	Slow	Real-time performance is insufficient

Comparison: Viola-Jones is fast enough for real-time operation but its accuracy is only 80%. Deep Learning provides 95% accuracy but is too slow according to the given conditions.

Evaluation: Neither method satisfies both constraints simultaneously.

Conclusion: Neither method should be selected directly. A lightweight Deep Learning model optimized for mobile devices should be used to achieve at least 85% accuracy while maintaining real-time performance.

10. Animation System Optimization

Requirements

- Realistic motion for cutscenes.
- Fast rendering for gameplay.

Approach	Realism	Speed
Motion Models	High	Low
Motion Estimation	Moderate	High

Comparison: Motion Models provide realistic movement but require more computation. Motion Estimation is faster but produces less realistic movement.

Evaluation: Cutscenes require realism, while gameplay requires fast response.

Conclusion: The system should use a **dual approach**. Motion Models should be used for cutscenes, while Motion Estimation should be used during gameplay.

11. Traffic Surveillance Model Selection

Requirements

- Accuracy at least 90%.
- Latency at most 80 ms.
- Must work under varying lighting.

Method	Accuracy	Latency	Evaluation
A	88%	60 ms	Accuracy is insufficient
B	91%	85 ms	Latency is too high
C	92%	75 ms	Satisfies accuracy and latency

Comparison: Method A has acceptable latency but insufficient accuracy. Method B has sufficient accuracy but exceeds the latency limit. Method C satisfies both numerical constraints.

Conclusion: Method C should be selected because it provides 92% accuracy and 75 ms latency. It should also be validated under different lighting conditions before deployment.

12. Drone Vision System Constraint

Requirements

- Lightweight computation.

- Real-time detection below 60 ms.
- Accuracy at least 85%.

Model	Accuracy	Latency	Evaluation
A	83%	40 ms	Accuracy is insufficient
B	87%	65 ms	Latency is too high
C	86%	55 ms	Satisfies requirements

Comparison: Model A is fast but does not reach the required accuracy. Model B has sufficient accuracy but exceeds the latency limit. Model C provides 86% accuracy and 55 ms latency.

Conclusion: **Model C is the most suitable choice** because it satisfies both the accuracy and real-time requirements while having moderate computational requirements.

13. Security System Recognition

Requirements

- Accuracy greater than 92%.
- Moderate computation.

Approach	Accuracy	Computation	Evaluation
Traditional	85%	Low	Accuracy is insufficient
Deep Learning	94%	High	Computation is excessive
Hybrid	91%	Moderate	Accuracy is insufficient

Comparison: Traditional methods have low computational requirements but insufficient accuracy. Deep Learning provides 94% accuracy but requires high computation. The Hybrid method has moderate computation but achieves only 91% accuracy.

Evaluation: None of the given approaches satisfies both requirements.

Conclusion: No available method completely satisfies the constraints. An optimized Deep Learning model should be developed using techniques such as pruning, quantization, or knowledge distillation to reduce computation while maintaining accuracy above 92%.

14. Motion Tracking in Sports Analytics

Requirements

- Accurate motion tracking.
- Real-time performance.

Method	Accuracy	Speed
Optical Flow	High	Slow
Motion Estimation	Lower	Fast

Comparison: Optical Flow provides accurate tracking but has high computational requirements. Motion Estimation is faster but sacrifices some accuracy.

Evaluation: The system requires both accuracy and real-time performance. Therefore, neither method alone provides the ideal solution.

Conclusion: A hybrid or optimized approach is recommended. Motion Estimation can be used for continuous real-time tracking, while Optical Flow can be applied selectively when higher accuracy is needed.

15. Face Recognition System Design

Requirements

- Accuracy at least 90%.
- Works in low light.

- Moderate computation.

Method	Accuracy	Computation	Evaluation
Traditional	82%	Low	Accuracy is insufficient
Deep Learning	93%	High	Accuracy is sufficient, computation is high
Hybrid	89%	Moderate	Accuracy is slightly insufficient

Comparison: Traditional methods require less computation but provide only 82% accuracy. The Hybrid approach has moderate computation but achieves only 89%, which is below the required 90%. Deep Learning provides 93% accuracy and is therefore the only listed approach that satisfies the accuracy requirement.

Evaluation: The main problem with Deep Learning is its high computational requirement. This can be reduced through model compression, pruning, quantization, and efficient architectures. Low-light image enhancement can also improve recognition performance.

Conclusion: Deep Learning is the best choice among the given options. It should be optimized to reduce computation and trained with low-light images to satisfy all deployment requirements.

14. Autonomous Navigation Constraint

Requirements

- Accuracy must be at least 95%.
- Latency must be less than 70 ms.

Model	Accuracy	Latency	Evaluation
A	94%	60 ms	Accuracy is insufficient
B	96%	80 ms	Latency is too high
C	95%	65 ms	Satisfies both constraints

Comparison: Model A has acceptable latency but its accuracy is below the required 95%. Model B provides high accuracy but its latency exceeds the 70 ms limit. Model C provides exactly 95% accuracy with 65 ms latency.

Conclusion: Model C should be selected because it satisfies both accuracy and latency requirements.

15. Industrial Inspection System

Requirements

- Defect detection accuracy must be at least 95%.
- Real-time operation is required.

Method	Accuracy	Speed	Evaluation
A	92%	Fast	Accuracy is insufficient
B	96%	Slow	Real-time requirement is not satisfied
C	95%	Moderate	Satisfies both requirements

Comparison: Method A is fast but its accuracy is only 92%. Method B provides 96% accuracy but is too slow for real-time operation. Method C provides exactly 95% accuracy with moderate processing speed.

Conclusion: Method C is the most suitable choice because it satisfies the required accuracy while maintaining practical processing speed.

16. Crowd Monitoring System

Requirements

- Stable detection in dense crowds.
- Moderate accuracy is acceptable.

Method	Dense Crowd Stability	Accuracy
Optical Flow	Unstable	Detailed
Motion Models	Stable	Moderate

Comparison: Optical Flow can provide detailed motion information, but overlapping movements in dense crowds can make its results unstable. Motion Models provide more stable tracking in crowded environments.

Evaluation: Stability is the primary requirement, while only moderate accuracy is needed. Therefore, the stability advantage of Motion Models is more important.

Conclusion: **Motion Models should be selected** because they provide stable detection in dense crowds and meet the required level of accuracy.

17. Video Analytics Pipeline

Requirements

- Accuracy must be at least 90%.
- Processing time must be less than 100 ms.

Pipeline	Accuracy	Processing Time	Evaluation
A	88%	80 ms	Accuracy is insufficient
B	91%	120 ms	Processing time is too high
C	90%	95 ms	Satisfies both constraints

Comparison: Pipeline A is fast but does not meet the accuracy requirement. Pipeline B provides high accuracy but exceeds the processing-time limit. Pipeline C provides 90% accuracy and processes within 95 ms.

Conclusion: **Pipeline C is the best choice** because it satisfies both constraints.

18. Smart Parking System

Requirements

- Accurate vehicle detection.
- Low computational cost.

Method	Accuracy	Computational Cost
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Viola-Jones	Low	Low
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YOLO	High	Moderate
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Comparison: Viola-Jones has low computational requirements, but its lower accuracy may result in incorrect vehicle detection. YOLO provides much better detection accuracy but requires moderate computation.

Evaluation: The system requires accurate vehicle detection as well as low computational cost. If the hardware can support moderate computation, YOLO provides a better practical balance.

Conclusion: **YOLO should be selected** when accurate vehicle detection is important. If the hardware is extremely limited, Viola-Jones can be considered, but detection accuracy will be lower.

19. Medical Imaging Analysis

Requirements

- Accuracy greater than 95%.
- Processing time is not critical.

Method	Accuracy	Processing Requirement
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Traditional	85%	Lower
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Deep Learning	97%	Higher
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Comparison: Traditional methods provide only 85% accuracy, which is far below the required level. Deep Learning provides 97% accuracy.

Evaluation: Since processing time is not critical, the computational cost of Deep Learning is acceptable. High accuracy is the primary requirement.

Conclusion: Deep Learning should be selected because it provides 97% accuracy and satisfies the requirement of greater than 95% accuracy.

20. Edge Device Vision System

Requirements

- Low power consumption.
- Real-time performance.

Model	Accuracy	Computation
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Deep Learning	High	Heavy
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Lightweight Model	Moderate	Efficient
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Comparison: Deep Learning provides high accuracy but requires significant processing power, which can increase energy consumption and reduce real-time performance on edge devices.

The Lightweight Model provides moderate accuracy with much lower computational requirements.

Evaluation: For an edge device, power consumption and response time are important. A lightweight model is therefore more suitable.

Conclusion: The Lightweight Model should be selected because it provides efficient computation, lower power consumption, and better real-time performance.

21. Traffic Flow Analysis

Requirements

- Detect motion patterns.
- Robust operation in crowded scenes.

Method	Motion Detail	Crowded-Scene Robustness
Optical Flow	Detailed	Noisy
Motion Estimation	Less detailed	Stable

Comparison: Optical Flow provides detailed motion information but can become noisy when many objects move simultaneously. Motion Estimation provides less detailed information but remains more stable.

Evaluation: Since the system specifically requires robustness in crowded scenes, stability is more important than highly detailed motion information.

Conclusion: **Motion Estimation should be selected** because it provides stable motion detection in crowded traffic conditions.

22. Retail Analytics System

Requirements

- Customer tracking.
- Moderate accuracy.
- Real-time processing.

Method	Accuracy	Speed
A	High	Slow
B	Moderate	Fast

Comparison: Method A provides better accuracy but is too slow for real-time customer tracking. Method B provides moderate accuracy and fast processing.

Evaluation: The system only requires moderate accuracy, while real-time processing is essential. Therefore, the speed advantage of Method B is more important.

Conclusion: Method B should be selected because it provides sufficient accuracy while maintaining real-time performance.

23. Object Detection in Foggy Conditions

Requirements

- Robust operation under poor visibility.
- Accuracy at least 85%.

Model	Accuracy	Speed
A	80%	Not specified
B	87%	Not specified
C	90%	Slow

Comparison: Model A does not meet the required 85% accuracy. Models B and C satisfy the accuracy requirement. Model C has the highest accuracy but is slow.

Evaluation: Since robustness under poor visibility is important, high accuracy is valuable. However, if real-time processing is also required, Model B may be more practical because Model C is slow.

Conclusion: Model B is the preferred balanced choice because it achieves 87% accuracy while avoiding the slow processing of Model C. If processing time is not important, Model C can be selected for maximum accuracy.

24. Video Surveillance Upgrade

Requirements

- Improve accuracy.

- Maintain the same latency.

Model	Accuracy	Latency
A	85%	50 ms
B	90%	50 ms

Comparison: Both models have exactly the same latency of 50 ms. However, Model B improves accuracy from 85% to 90%.

Evaluation: Since the objective is to improve accuracy without increasing latency, Model B clearly provides a better result.

Conclusion: Model B should be selected because it increases accuracy to 90% while maintaining the same 50 ms latency.

25. Robotics Vision System

Requirements

- Fast decision-making.
- Moderate accuracy.

Method	Speed	Accuracy
Deep Learning	Slow	High
Traditional	Fast	Moderate

Comparison: Deep Learning provides high accuracy but requires more processing time. Traditional computer vision methods are faster and provide moderate accuracy.

Evaluation: The system prioritizes fast decision-making, while only moderate accuracy is required. Therefore, the speed advantage of Traditional methods is more important.

Conclusion: Traditional methods should be selected because they provide fast decision-making while meeting the moderate accuracy requirement.

26. Smart Home Security

Requirements

- Accurate intrusion detection.
- Low computation.

Method	Accuracy	Computation
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YOLO	High	Moderate
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Viola-Jones	Lower	Low
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Comparison: YOLO provides better intrusion detection accuracy but requires more computational resources. Viola-Jones is computationally efficient but may produce more detection errors.

Evaluation: The system requires both accuracy and low computation. If the smart-home device has moderate processing capability, YOLO provides the better balance. For extremely low-power hardware, Viola-Jones can be used with reduced accuracy.

Conclusion: **YOLO should be preferred** when accurate intrusion detection is the main priority and moderate computation is available.

27. Autonomous Drone Tracking

Requirements

- Real-time response below 40 ms.
- Accuracy at least 85%.

Model	Accuracy	Latency	Evaluation
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A	83%	30 ms	Accuracy is insufficient
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Model	Accuracy	Latency	Evaluation
B	86%	50 ms	Latency is too high
C	88%	38 ms	Satisfies both constraints

Comparison: Model A is fast but its accuracy is below 85%. Model B provides sufficient accuracy but has 50 ms latency, exceeding the 40 ms limit. Model C provides 88% accuracy with only 38 ms latency.

Conclusion: Model C should be selected because it satisfies both the accuracy and real-time requirements.

28. Motion Animation System

Requirements

- Smooth motion.
- Real-time rendering.

Method	Realism/Smoothness	Speed
Motion Models	High	Slow
Motion Estimation	Moderate	Fast

Comparison: Motion Models provide realistic and smooth motion but require high computational resources. Motion Estimation is faster but provides less realistic movement.

Evaluation: Real-time rendering requires efficient computation. Therefore, using Motion Models continuously may reduce rendering performance.

Conclusion: Motion Estimation should be used for real-time rendering because it provides faster processing. Motion Models can be used selectively for important scenes where higher realism is required.

Code of Conduct:

I K.L.V.S KARTHIK (192424177) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

K.L.V.S KARTHIK

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution

Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough	Good	Basic	Poor or no

		analysis with validated results and performance evaluation	analysis with reasonable validation	analysis with limited validation	analysis
Professionalism	5%	Highly professional, well documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation